SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining
COURSE CODE : CSA1610

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I P.Poojitha(192472349) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Poojitha

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	**Claims per 100 Policyholders (yi)**
3.2	15
4.0	18
5.5	25
4.8	20

SOLUTION:

To determine whether the premium increase rate and claim frequency have a positive or inverse relationship using covariance.
Given Data

Premium Increase (x)	**Claims (y)**
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate Mean

Step 2: Covariance Table

x	**y**	**$x-\bar{x}$**	**$y-\bar{y}$**	**$(x-\bar{x})(y-\bar{y})$**
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Sum = **12.25**

Step 3: Covariance

Result

Covariance = 3.0625
Since the covariance is **positive**, premium increase and claim frequency have a **positive relationship**.

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

SOLUTION:

Given Data
110,112,115,115,118,120,120,122,124,124,126,126,126,126,130,135,135,138,138,138,138,140,145,150,152,160,180
Total values = **27**
Divide into **3 equal-frequency bins (9 values each).**

Bin 1
110,112,115,115,118,120,120,122,124
Mean

Smoothed Values
117.33,117.33,117.33,117.33,117.33,117.33,117.33,117.33,117.33

Bin 2
124,126,126,126,126,130,135,135,138
Mean

Smoothed Values
129.56 repeated 9 times.

Bin 3
138,138,140,145,150,152,160,180
Mean

Smoothed Values
150.38 repeated 8 times.

Result
Equal-frequency binning smooths the blood pressure readings by replacing values with the **bin mean**, reducing minor fluctuations.

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

SOLUTION:

Given BMI Data
18.5,19.2,21.8,24.9,25.5,26.2,26.4,27.8,29.2,30.2,30.4,31.9,32.4,33.1,33.6,34.7,38.2,39.5

Step 1
Minimum = 18.5
Maximum = 39.5
Range

Number of bins = 3
Bin Width

Bins
Bin 1: 18.5 – 25.5
18.5,19.2,21.8,24.9,25.5
Boundary Values
18.5,18.5,18.5,25.5,25.5

Bin 2: 25.5 – 32.5
26.2,26.4,27.8,29.2,30.2,30.4,31.9,32.4
Boundary Values
25.5,25.5,25.5,32.5,32.5,32.5,32.5,32.5

Bin 3: 32.5 – 39.5
33.1,33.6,34.7,38.2,39.5
Boundary Values
32.5,32.5,32.5,39.5,39.5

Result
Equal-width binning classifies BMI values into risk groups using fixed-width intervals.

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.

To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

SOLUTION:

Given Data
120,140,145,145,148,150,150,150,155,155,155,155,160,160,165,170,170,175,175,175,175,180,190,200,210,230,300

Bin 1
120,140,145,145,148,150,150,150,155
Median = **148**
New Values
148,148,148,148,148,148,148,148,148

Bin 2
155,155,155,160,160,165,170,170,175
Median = **160**
New Values
160 repeated 9 times.

Bin 3
175,175,175,180,190,200,210,230,300
Median = **190**
New Values
190 repeated 9 times.

Result
Replacing each value with the **bin median** reduces data size while preserving the central tendency.

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

SOLUTION:

Given Ages
18,22,25,42,28,43,33,35,56,28

Step 1: Mean

Step 2: Mean Absolute Deviation (MAD)

Step 3: Standard Deviation

Step 4: Z-score
Formula

Age	Z-score
18	−1.40
22	−1.03
25	−0.75
42	0.84

Result

- **Mean = 33**
- **MAD = 8.8**
- **Z-scores**
  - 18 → **−1.40**
  - 22 → **−1.03**
  - 25 → **−0.75**
  - 42 → **0.84**